

Virtual Private Cloud

A VPC is like your personal, isolated section of AWS cloud where you have full control over networking.

- The overall IP range of the VPC.
10.0.0.0/16 (gives 65,536 private IPs)

1. Create VPC

The screenshot shows the 'Create VPC' page in the AWS Management Console. The breadcrumb trail is 'VPC > Your VPCs > Create VPC'. The page title is 'Create VPC' with an 'Info' link. A description states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.' The 'VPC settings' section includes 'Resources to create' with 'VPC only' selected, 'Name tag - optional' with 'Practice' entered, 'IPv4 CIDR block' with 'IPv4 CIDR manual input' selected and '172.25.0.0/16' entered, and 'IPv6 CIDR block' with 'No IPv6 CIDR block' selected. The console header shows the AWS logo, a search bar, account ID '0950-7410-7161', and region 'Europe (Stockholm)'.

2. Create Subnet

You divide the VPC into smaller networks.

- **Public Subnets** → can access internet
- **Private Subnets** → no direct internet access

The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb trail is 'VPC > Subnets > Create subnet'. The page title is 'Subnet settings' with a description: 'Specify the CIDR blocks and Availability Zone for the subnet.' Under 'Subnet 1 of 1', the 'Subnet name' is 'my-subnet-1'. The 'Availability Zone' is set to 'No preference'. The 'IPv4 VPC CIDR block' is '172.25.0.0/16'. The 'IPv4 subnet CIDR block' is '172.25.0.0/26', which provides '4,096 IPs'. The 'Tags - optional' section shows a key 'Name' with value 'my-subnet-1'. The console header is identical to the previous screenshot.

aws [Search] [Alt+S] Europe (Stockholm) Account ID: 0950-7410-7161 Sakshi Jain

VPC > Subnets > Create subnet

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
my-subnet-2
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
172.25.0.0/16

IPv4 subnet CIDR block
172.25.16.0/20 4,096 IPs

▼ **Tags - optional**

Key	Value - optional	
Q Name	Q my-subnet-2	Remove

Add new tag
You can add 49 more tags.

Here I can create **Maximum 16** subnet because my **CIDR is 16**

Subnets (2) Info

Last updated less than a minute ago Actions Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-042b5c00182e69732 Subnet ID : subnet-05829f57c0ece6527 Clear filters

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	my-subnet-1	subnet-042b5c00182e69732	Available	vpc-0bca94b577e49b72c Prac...	Off	172.25.0.0/20
☐	my-subnet-2	subnet-05829f57c0ece6527	Available	vpc-0bca94b577e49b72c Prac...	Off	172.25.16.0/20

Subnets (1/2) Info

Last updated 2 minutes ago Actions Create subnet

Find subnets by attribute or tag

my-subnet-1 Clear filters

☒	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☒	my-subnet-1	subnet-042b5c00182e69732	Available	vpc-0bca94b577e49b72c Prac...	Off	172.25.0.0/20
☐	my-subnet-2	subnet-05829f57c0ece6527	Available	vpc-0bca94b577e49b72c Prac...	Off	172.25.16.0/20

- View details
- Create flow log
- Edit subnet settings
- Edit IPv6 CIDRs
- Edit network ACL association
- Edit route table association
- Edit CIDR reservations
- Share subnet
- Manage tags
- Delete subnet

subnet-042b5c00182e69732 / my-subnet-1

Details | Flow logs | Route table | Network ACL | CIDR reservations | Share | Tags

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

VPC > Subnets > subnet-042b5c00182e69732 > Edit subnet settings

Edit subnet settings [Info](#)

Subnet

Subnet ID

subnet-042b5c00182e69732

Name

my-subnet-1

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch [Info](#)

☐ Enable resource name DNS AAAA record on launch [Info](#)

Hostname type [Info](#)

3. Internet Gateway

An **Internet Gateway (IGW)** is a **VPC component** that allows your resources (like EC2 instances) to: Access the internet, Be accessed from the internet.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

IGW-1

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional
Name	IGW-1

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. For 1 Internet Gateway we can attach single VPC Only.

Internet gateways (1/2) [Info](#)

[Find internet gateways by attribute or tag](#)

	Name	Internet gateway ID	State	VPC ID
☐	-	igw-07a534c010dad6430	Attached	vpc-09b6bff8ffa12c928
☒	IGW-1	igw-01025daca8f434ddb	Detached	-

Actions [Create internet gateway](#)

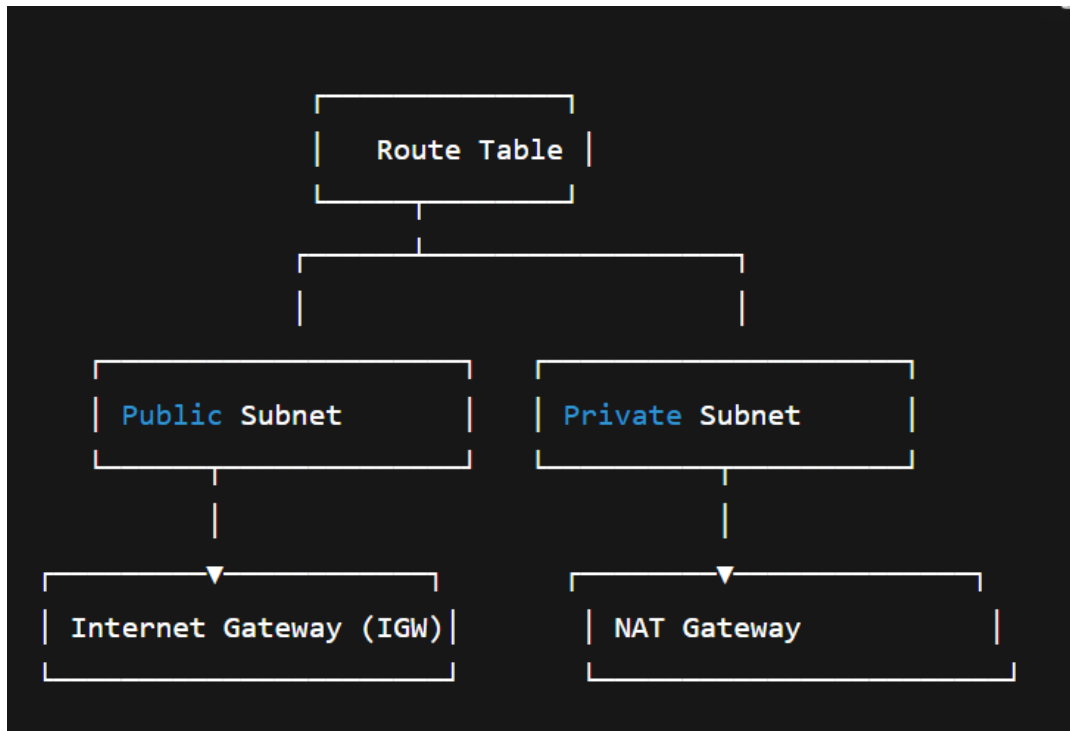
- View details
- Attach to VPC
- Detach from VPC
- Manage tags
- Delete internet gateway

Info

Cancel Attach internet gateway

4. Route Tables

A **Route Table** is a set of rules that decide **where your network traffic should go** inside a VPC.



aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 0950-7410-7161

Sakshi Jain

VPC > Route tables > Create route table

Info

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

my-route-table

VPC
The VPC to use for this route table.

vpc-0bca94b577e49b72c (Practice)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Q Name	Q my-route-table	Remove
Add new tag		

You can add 49 more tags.

[Cancel](#) [Create route table](#)

Click on Edit Routes

Route tables (1/3)

Info

Last updated less than a minute ago

Actions

Create route table

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Ow
☐	-	rtb-0a76891d23cfab0df	-	-	Yes	vpc-09b6bff8ffa12c928	095
☐	-	rtb-064731d9f28656ad4	-	-	Yes	vpc-0bca94b577e49b72c Prac...	095
☒	my-route-table	rtb-07b653b201fdc6fda	-	-	No	vpc-0bca94b577e49b72c Prac...	095

rtb-07b653b201fdc6fda / my-route-table

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
172.25.0.0/16	local	Active	No	Create Route Table

VPC > Route tables > rtb-07b653b201fdc6fda > Edit routes

Edit routes

Destination

172.25.0.0/16

Target

local

Q local

Internet Gateway

igw-01025daca8f434ddb

Status

Active

-

Propagated

No

No

Route Origin

CreateRouteTable

CreateRoute

Add route

Remove

Cancel

Preview

Save changes

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-01025daca8f434ddb	Active	No	Create Route
172.25.0.0/16	local	Active	No	Create Route Table

Route tables (1/3)

Info

Last updated less than a minute ago

Actions

Create route table

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Ow
☐	-	rtb-0a76891d23cfab0df	-	-	Yes	vpc-09b6bff8ffa12c928	095
☐	-	rtb-064731d9f28656ad4	-	-	Yes	vpc-0bca94b577e49b72c Prac...	095
☒	my-route-table	rtb-07b653b201fdc6fda	2 subnets	-	No	vpc-0bca94b577e49b72c Prac...	095

View details

Set main route table

Edit subnet associations

Edit edge associations

Edit route propagation

Edit routes

Manage tags

Delete route table

rtb-07b653b201fdc6fda / my-route-table

Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (2/2)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
my-subnet-1	subnet-042b5c00182e69732	172.25.0.0/20	-	Main (rtb-064731d9f28656ad4)
my-subnet-2	subnet-05829f57c0ece6527	172.25.16.0/20	-	Main (rtb-064731d9f28656ad4)

Selected subnets

subnet-05829f57c0ece6527 / my-subnet-2 subnet-042b5c00182e69732 / my-subnet-1

[Cancel](#) [Save associations](#)

5. Create Instance

Auto-Assign Public IP should be enabled so, that you use instance with public IP

Launch an instance

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
Practice [Create new key pair](#)

Network settings

VPC - required
vpc-0bca94b577e49b72c (Practice)

Subnet
subnet-042b5c00182e69732 my-subnet-1
VPC: vpc-0bca94b577e49b72c Owner: 095074107161 Availability Zone: eu-north-1a (eu-n1-az1) Zone type: Availability Zone IP addresses available: 4091 CIDR: 172.25.0.0/20

Auto-assign public IP
Enable

Firewall (security groups)
Create security group ☒ Select existing security group ☐

Security group name - required
launch-wizard-3

Summary

Number of instances: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-0f50f13aefb6c0a5d

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Instances (1/1)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) [Running](#)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
vpc-test	i-04128f90c2e7cd8b7	Running	t3.micro	Initializing	View alarms +	eu-north-1a	-

i-04128f90c2e7cd8b7 (vpc-test)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary

Instance ID
i-04128f90c2e7cd8b7

Public IPv4 address
51.20.60.27 | [open address](#)

Private IPv4 addresses
172.25.0.144

